

Zidane Ho

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EDUCATION

UC San Diego

La Jolla, CA

Bachelors of Science in Computer Science

Expected June 2028

- **Relevant Coursework:** System Programming and Software Tools, Computer Organization, Graph Theory, Mathematical Reasoning, Advanced Data Structures, Design and Analysis of Algorithms, Intro to Computer Graphics, Intro to ML, Theory of Computability
- **GPA:** 3.7/4.0

TECHNICAL SKILLS

Languages: Python, C#, C++, Java, TypeScript, JavaScript, C, HTML/CSS, R

Frameworks & Libraries: React, Next.js, Unity, Godot, FastAPI, Vite, Tailwind CSS, Three.js, sklearn, Git, Docker, Postman, ASP.NET Core, Solid.js, Playroomkit

EXPERIENCE

IT Services at UCSD

February 2026 - Present

Student Floating Engineer

C#, Docker, Kubernetes

- Manage university infrastructure and verify student course access using command-line operations, Docker, and Kubernetes.
- Resolve technical support tickets by troubleshooting and debugging user configurations within Jupyter Notebook container environments.

ACM AI at UCSD

October 2025 - June 2026

Competitions Staff

python, pygame

- Collaborated with ACM AI's board team to develop AI-related competitions for the UCSD community.
- Developed the GUI for the competition Stellatro primarily using pygame. [Github Link](#)

AWARDS

Blockography.AI - Minecraft Biome Classifier | *Python, scikit-image, OpenCV*

November 2025

- Achieved 1st on the beginner track, 5th out of 45 teams overall, in UCSD ACM's Blockography.AI competition by developing a high performance XGBooster model in Python that classifies Minecraft biomes from a dataset of screenshots in the game.
- Optimized the model's test accuracy from 60% to 92% through tuning hyperparameters, resizing images, and feature engineering. [GitHub Link](#)

PROJECTS

SamiraMod : A Risk of Rain 2 Mod | *C#, Unity*

July 2024 - August 2024

- Engineered and launched a character mod, achieving over **170K+ downloads** on Thunderstore.io [Link](#)
- Developed custom character abilities, animations, and gameplay mechanics using Risk of Rain 2's API and C#.
- Iteratively improved the mod by actively managing user feedback and resolving issues via the project's repository. [GitHub Link](#).

Lab 30: TritonWare 2024 | *C#, Unity*

October 2024

- Led a 5-member team to design and deliver a top-down horror game at UCSD's TritonWare kickoff.
- Built core gameplay loop in C#, including movement, sprint, flashlight, and UI systems.
- Developed enemy AI using A Pathfinding and FSMs for pursuit and attack behaviors. [GitHub Link](#)

Doodle Derby | *HTML, CSS, Typescript, Solid.js, Konva.js, Playroomkit*

February 2026

- Collaborated in a cross-functional team of 15, translating artistic assets and sound design into a cohesive, low-latency frontend experience.

- Architected a real-time multiplayer lobby using Playroomkit, enabling seamless synchronization of game settings and player states between host and clients.
- Engineered a responsive web-based gaming interface using Solid.js, resulting in a seamless user experience across desktop platforms. [GitHub Link](#) [Website Link](#)

Auto Battler Game Engine | *Typescript, React, Tailwind CSS, Next.js, Three.js* *February 2025 - Present*

- Developed a rogue-like auto battler web game, engineering a game engine by using Three.js and React.
- Implemented a scalable game loop that contains dynamic environments, unique character ability, and scalable difficulty. [Project Link](#)

Steam Analysis : A Sentiment Analysis | *Python, pandas, scikit-learn* *March 2025 - August 2025*

- Created a full-stack research tool to provide game developers with competitive analysis of similar games on Steam.
- Utilized cosine similarity on review embeddings to identify and rank analogous games for market research.
- Integrated the Gemini API to perform sentiment analysis and generate concise summaries of thousands of user reviews. [GitHub Link](#)

Spotlight : CalHacks 2025 | *Chrome Extension API (MV3), Claude API, FishAudio API* *October 2025*

- Developed a voice-guided web navigation assistant to help users, especially the elderly or those with cognitive challenges, browse websites using voice commands and visual guidance.
- Integrated the Claude AI API to understand complex, natural language commands; the AI reasons over a DOM snapshot and screenshot to identify the correct element to interact with.
- Engineered a multi-layer visual feedback system (smart highlighting, arrow pointers, animated cursor, and tooltips) to guide the user to the target element, prioritizing user control over full automation.

LectureLens : DiamondHacks 2026 | *React, FastAPI, Gemini API, TwelveLabs* *April 2026*

- Developed an AI-powered educational platform that automatically processes lecture videos into structured study materials, comprehensive notes, and interactive timelines.
- Integrated TwelveLabs and Google Gemini APIs for multi-modal analysis, extracting key concepts and generating persona-driven, customizable notes from both visual and audio content.
- Architected an asynchronous processing pipeline using FastAPI, Celery, Redis, and MongoDB, paired with a React frontend featuring context-aware Q&A and precise timestamp navigation. [Devpost Link](#)